Trabalho Final - CE442A

Eduardo Ranieri, Gabriel Lucena, William Makino

Profº Marcelo Justus

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Introdução

O presente trabalho busca entender o efeito da *Covid-19* sobre a arrecadação federal, especialmente o começo do *isolamento social* em março/2020. É usada a Metodologia Box-Jenkins, com a função *auto.arima* para fazer uma *análise de intervenção* na série temporal selecionada. O objetivo da análise de intervenção é conferir se uma série temporal foi afetada graças a um evento exógeno.

1ª Etapa: Prever a Série - Pré Intervenção

Afim de fazer a previsão da série, sem *a priori* realizar a análise de intervenção, é necessário usar a função *auto.arima*. Essa etapa é importante para encontrarmos um modelo que descreva o processo estocástico da série. Conseguindo descrever esse processo, é possível testar o impacto da intervenção exógena sobre a série temporal. Para tanto, seguimos os passos:

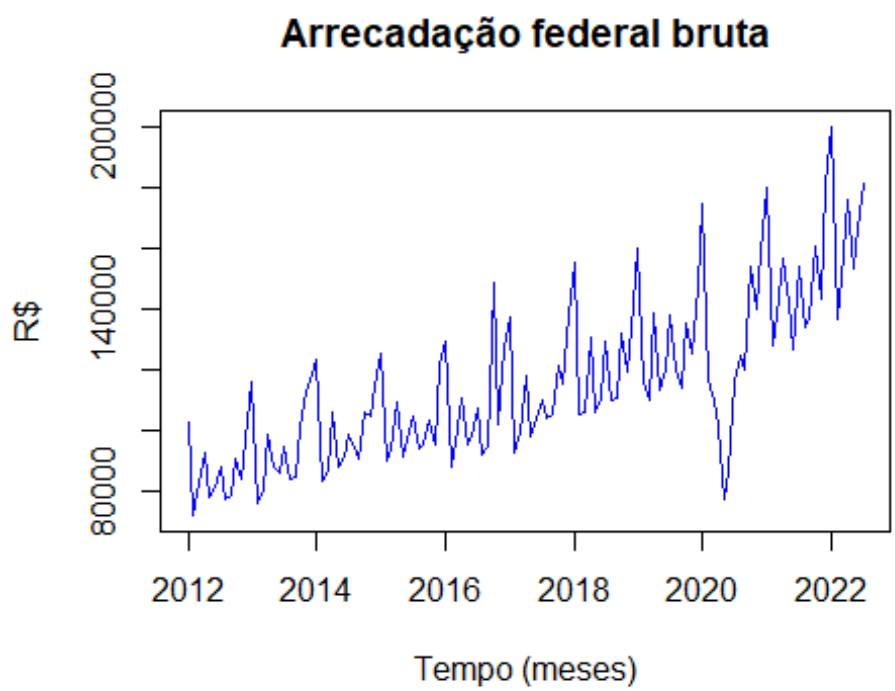
Transformação dos dados em Série Temporal usando o pacote "forecast"

```
# Carregando os pacotes instalados
require(readxl)

## Carregando pacotes exigidos: readxl

## Warning: package 'readxl' was built under R version 4.2.3

require(forecast)

## Carregando pacotes exigidos: forecast

## Warning: package 'forecast' was built under R version 4.2.3

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo

require(FinTS)

## Carregando pacotes exigidos: FinTS

## Warning: package 'FinTS' was built under R version 4.2.3

## Carregando pacotes exigidos: zoo

## Warning: package 'zoo' was built under R version 4.2.3

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric

##
## Attaching package: 'FinTS'

## The following object is masked from 'package:forecast':
##
##     Acf

library(tsoutliers)

## Warning: package 'tsoutliers' was built under R version 4.2.3
```

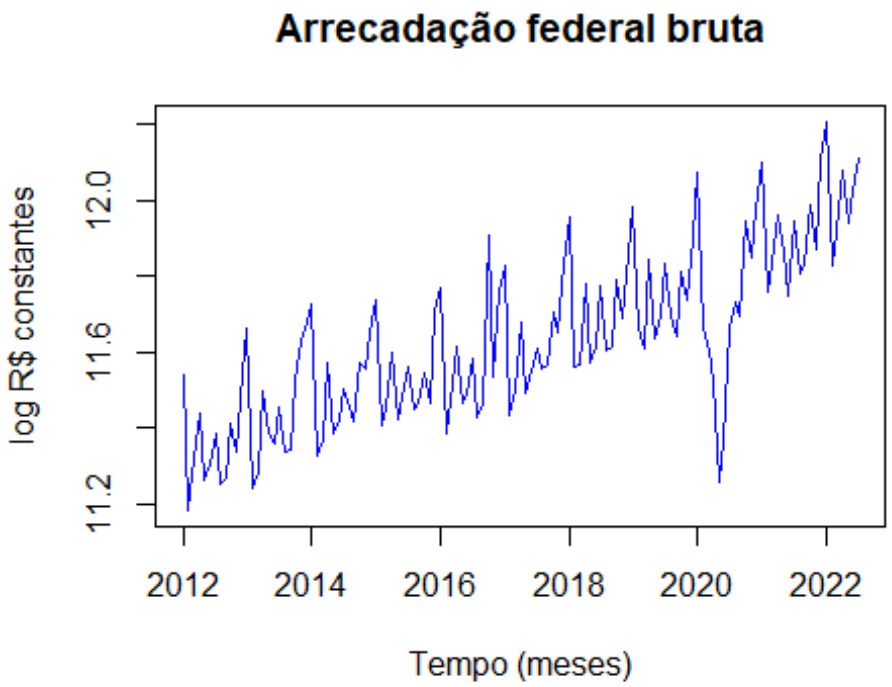
```
#Lendo o arquivo - dados estão na variável arrecadacao
arrecadacao <- read_excel("arrecadacao.xlsx")

## New names:
## • `` -> `...5`

# Transformando o dado em um objeto ts
arrecad_ts <- ts(arrecadacao[-1,3],
                  start=c(2012,1), frequency=12)
```

Plot da série

```
# Plotando a série
plot.ts(arrecad_ts, xlab="Tempo (meses)",
        ylab='R$',
        main="Arrecadação federal bruta",
        col="blue",
        type="l")
```



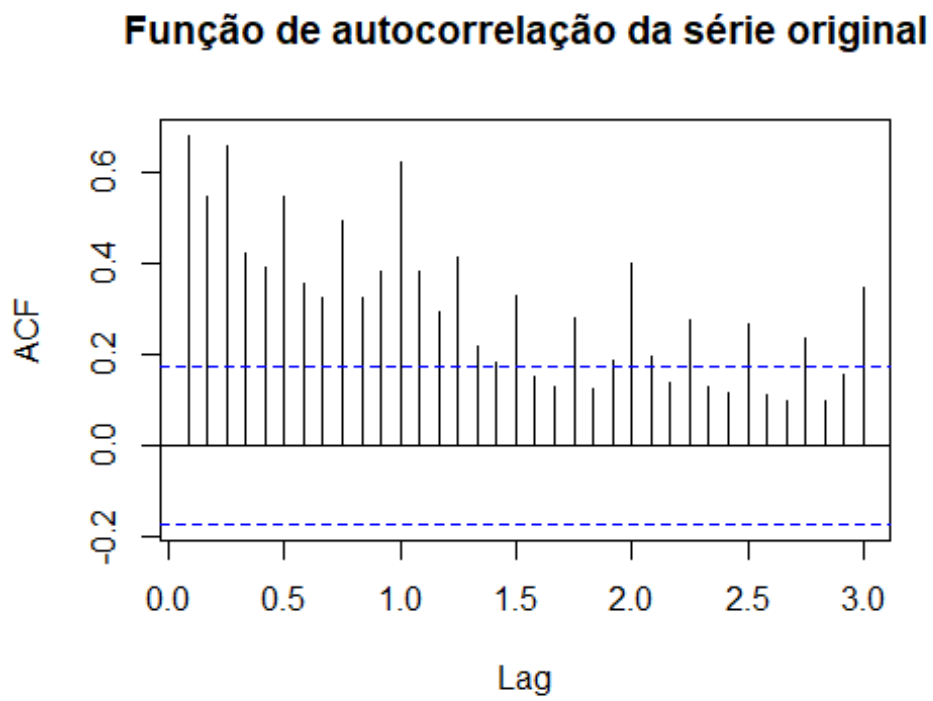

```
# Plotando o log da série
plot.ts(log(arrecad_ts), xlab="Tempo (meses)",
        ylab='log R$,
        main="Arrecadação federal bruta",
        col="blue",
        type="l")
```



Podemos observar pela representação gráfica que a série tem uma tendência clara e possivelmente alguns pontos de sazonalidade. Pelo log da série, feito para diminuir a variabilidade, podemos ver esses componentes também.

Estimando a Função de Autocorrelação (FAC)

```
# Estimando a função de autocorrelação (FAC)
Acf(log(arrecad_ts), lag.max=36,
    main= "Função de autocorrelação da série original")
```



A estimação da FAC serve para avaliar a estacionariedade da série e a ordem da parte autoregressiva do modelo. Ao observar a FAC dessa série, não se pode concluir sobre sua estacionariedade, visto que a maioria das colunas em lag ultrapassa o limite de significância, o que pode indicar autocorrelação, ainda que eventualmente elas entrem no limite de significância. Além disso, pela análise visual feita anteriormente ao plotar o gráfico da série temporal (normal e em log), percebemos componentes sazonais e tendência. Portanto, julgamos não ser necessário realizar um teste de Raiz Unitária.

Identificando um modelo candidato

```
# Primeiro será escolhido um modelo candidato, utilizando a seleção
automática com busca longa sugerida pela função auto.arima

mod_arrecad <- auto.arima(arrecad_ts, lambda=0,
                      stepwise=FALSE, trace=TRUE,
                      approximation=FALSE, allowdrift=TRUE,
                      allowmean=TRUE, ic=c("aicc"))

##
##  ARIMA(0,0,0)(0,1,0)[12]                    : -129.9823
##  ARIMA(0,0,0)(0,1,0)[12] with drift         : -157.5744
##  ARIMA(0,0,0)(0,1,1)[12]                    : -129.3778
##  ARIMA(0,0,0)(0,1,1)[12] with drift         : Inf
##  ARIMA(0,0,0)(0,1,2)[12]                    : -127.3986
##  ARIMA(0,0,0)(0,1,2)[12] with drift         : Inf
##  ARIMA(0,0,0)(1,1,0)[12]                    : -129.3783
##  ARIMA(0,0,0)(1,1,0)[12] with drift         : -172.2645
```

```
##  ARIMA(0,0,0)(1,1,1)[12]                     : -127.2934
##  ARIMA(0,0,0)(1,1,1)[12] with drift          : Inf
##  ARIMA(0,0,0)(1,1,2)[12]                     : Inf
##  ARIMA(0,0,0)(1,1,2)[12] with drift          : Inf
##  ARIMA(0,0,0)(2,1,0)[12]                     : -127.4515
##  ARIMA(0,0,0)(2,1,0)[12] with drift          : -203.0806
##  ARIMA(0,0,0)(2,1,1)[12]                     : Inf
##  ARIMA(0,0,0)(2,1,1)[12] with drift          : -214.9404
##  ARIMA(0,0,0)(2,1,2)[12]                     : Inf
##  ARIMA(0,0,0)(2,1,2)[12] with drift          : -212.7133
##  ARIMA(0,0,1)(0,1,0)[12]                     : -183.2709
##  ARIMA(0,0,1)(0,1,0)[12] with drift          : -199.9541
##  ARIMA(0,0,1)(0,1,1)[12]                     : -188.0205
##  ARIMA(0,0,1)(0,1,1)[12] with drift          : Inf
##  ARIMA(0,0,1)(0,1,2)[12]                     : -186.0571
##  ARIMA(0,0,1)(0,1,2)[12] with drift          : Inf
##  ARIMA(0,0,1)(1,1,0)[12]                     : -187.5384
##  ARIMA(0,0,1)(1,1,0)[12] with drift          : -217.372
##  ARIMA(0,0,1)(1,1,1)[12]                     : -185.951
##  ARIMA(0,0,1)(1,1,1)[12] with drift          : Inf
##  ARIMA(0,0,1)(1,1,2)[12]                     : Inf
##  ARIMA(0,0,1)(1,1,2)[12] with drift          : Inf
##  ARIMA(0,0,1)(2,1,0)[12]                     : -186.6681
##  ARIMA(0,0,1)(2,1,0)[12] with drift          : -235.4647
##  ARIMA(0,0,1)(2,1,1)[12]                     : Inf
##  ARIMA(0,0,1)(2,1,1)[12] with drift          : Inf
##  ARIMA(0,0,1)(2,1,2)[12]                     : Inf
##  ARIMA(0,0,1)(2,1,2)[12] with drift          : Inf
##  ARIMA(0,0,2)(0,1,0)[12]                     : -197.6788
##  ARIMA(0,0,2)(0,1,0)[12] with drift          : -209.6286
##  ARIMA(0,0,2)(0,1,1)[12]                     : -207.735
##  ARIMA(0,0,2)(0,1,1)[12] with drift          : Inf
##  ARIMA(0,0,2)(0,1,2)[12]                     : -205.6893
##  ARIMA(0,0,2)(0,1,2)[12] with drift          : Inf
##  ARIMA(0,0,2)(1,1,0)[12]                     : -205.7159
##  ARIMA(0,0,2)(1,1,0)[12] with drift          : -228.6868
##  ARIMA(0,0,2)(1,1,1)[12]                     : -205.618
##  ARIMA(0,0,2)(1,1,1)[12] with drift          : Inf
##  ARIMA(0,0,2)(1,1,2)[12]                     : Inf
##  ARIMA(0,0,2)(1,1,2)[12] with drift          : Inf
##  ARIMA(0,0,2)(2,1,0)[12]                     : -206.924
##  ARIMA(0,0,2)(2,1,0)[12] with drift          : -244.1084
##  ARIMA(0,0,2)(2,1,1)[12]                     : -204.7083
##  ARIMA(0,0,2)(2,1,1)[12] with drift          : Inf
##  ARIMA(0,0,3)(0,1,0)[12]                     : -209.3739
##  ARIMA(0,0,3)(0,1,0)[12] with drift          : -216.5574
##  ARIMA(0,0,3)(0,1,1)[12]                     : -225.8534
##  ARIMA(0,0,3)(0,1,1)[12] with drift          : Inf
##  ARIMA(0,0,3)(0,1,2)[12]                     : -223.864
##  ARIMA(0,0,3)(0,1,2)[12] with drift          : Inf
```

```
##  ARIMA(0,0,3)(1,1,0)[12]                     : -222.66
##  ARIMA(0,0,3)(1,1,0)[12] with drift          : -238.3625
##  ARIMA(0,0,3)(1,1,1)[12]                     : -223.779
##  ARIMA(0,0,3)(1,1,1)[12] with drift          : Inf
##  ARIMA(0,0,3)(2,1,0)[12]                     : -224.1405
##  ARIMA(0,0,3)(2,1,0)[12] with drift          : -250.5469
##  ARIMA(0,0,4)(0,1,0)[12]                     : -207.6044
##  ARIMA(0,0,4)(0,1,0)[12] with drift          : -214.3391
##  ARIMA(0,0,4)(0,1,1)[12]                     : -227.7309
##  ARIMA(0,0,4)(0,1,1)[12] with drift          : Inf
##  ARIMA(0,0,4)(1,1,0)[12]                     : -222.1683
##  ARIMA(0,0,4)(1,1,0)[12] with drift          : -236.245
##  ARIMA(0,0,5)(0,1,0)[12]                     : -205.7498
##  ARIMA(0,0,5)(0,1,0)[12] with drift          : -212.0955
##  ARIMA(1,0,0)(0,1,0)[12]                     : -213.547
##  ARIMA(1,0,0)(0,1,0)[12] with drift          : -217.6728
##  ARIMA(1,0,0)(0,1,1)[12]                     : Inf
##  ARIMA(1,0,0)(0,1,1)[12] with drift          : Inf
##  ARIMA(1,0,0)(0,1,2)[12]                     : Inf
##  ARIMA(1,0,0)(0,1,2)[12] with drift          : Inf
##  ARIMA(1,0,0)(1,1,0)[12]                     : -232.6097
##  ARIMA(1,0,0)(1,1,0)[12] with drift          : -239.7804
##  ARIMA(1,0,0)(1,1,1)[12]                     : Inf
##  ARIMA(1,0,0)(1,1,1)[12] with drift          : Inf
##  ARIMA(1,0,0)(1,1,2)[12]                     : Inf
##  ARIMA(1,0,0)(1,1,2)[12] with drift          : Inf
##  ARIMA(1,0,0)(2,1,0)[12]                     : -240.3894
##  ARIMA(1,0,0)(2,1,0)[12] with drift          : -252.5223
##  ARIMA(1,0,0)(2,1,1)[12]                     : -248.7133
##  ARIMA(1,0,0)(2,1,1)[12] with drift          : Inf
##  ARIMA(1,0,0)(2,1,2)[12]                     : Inf
##  ARIMA(1,0,0)(2,1,2)[12] with drift          : Inf
##  ARIMA(1,0,1)(0,1,0)[12]                     : -211.584
##  ARIMA(1,0,1)(0,1,0)[12] with drift          : -215.5303
##  ARIMA(1,0,1)(0,1,1)[12]                     : Inf
##  ARIMA(1,0,1)(0,1,1)[12] with drift          : Inf
##  ARIMA(1,0,1)(0,1,2)[12]                     : Inf
##  ARIMA(1,0,1)(0,1,2)[12] with drift          : Inf
##  ARIMA(1,0,1)(1,1,0)[12]                     : -231.7256
##  ARIMA(1,0,1)(1,1,0)[12] with drift          : -237.7025
##  ARIMA(1,0,1)(1,1,1)[12]                     : Inf
##  ARIMA(1,0,1)(1,1,1)[12] with drift          : Inf
##  ARIMA(1,0,1)(1,1,2)[12]                     : Inf
##  ARIMA(1,0,1)(1,1,2)[12] with drift          : Inf
##  ARIMA(1,0,1)(2,1,0)[12]                     : -241.2174
##  ARIMA(1,0,1)(2,1,0)[12] with drift          : -250.505
##  ARIMA(1,0,1)(2,1,1)[12]                     : Inf
##  ARIMA(1,0,1)(2,1,1)[12] with drift          : Inf
##  ARIMA(1,0,2)(0,1,0)[12]                     : -209.708
##  ARIMA(1,0,2)(0,1,0)[12] with drift          : -214.2194
```

```
##  ARIMA(1,0,2)(0,1,1)[12]                     : Inf
##  ARIMA(1,0,2)(0,1,1)[12] with drift          : Inf
##  ARIMA(1,0,2)(0,1,2)[12]                     : Inf
##  ARIMA(1,0,2)(0,1,2)[12] with drift          : Inf
##  ARIMA(1,0,2)(1,1,0)[12]                     : -229.5388
##  ARIMA(1,0,2)(1,1,0)[12] with drift          : -236.3911
##  ARIMA(1,0,2)(1,1,1)[12]                     : Inf
##  ARIMA(1,0,2)(1,1,1)[12] with drift          : Inf
##  ARIMA(1,0,2)(2,1,0)[12]                     : -240.336
##  ARIMA(1,0,2)(2,1,0)[12] with drift          : -248.9019
##  ARIMA(1,0,3)(0,1,0)[12]                     : -208.0161
##  ARIMA(1,0,3)(0,1,0)[12] with drift          : -214.3383
##  ARIMA(1,0,3)(0,1,1)[12]                     : Inf
##  ARIMA(1,0,3)(0,1,1)[12] with drift          : Inf
##  ARIMA(1,0,3)(1,1,0)[12]                     : -228.468
##  ARIMA(1,0,3)(1,1,0)[12] with drift          : -236.29
##  ARIMA(1,0,4)(0,1,0)[12]                     : Inf
##  ARIMA(1,0,4)(0,1,0)[12] with drift          : Inf
##  ARIMA(2,0,0)(0,1,0)[12]                     : -211.5988
##  ARIMA(2,0,0)(0,1,0)[12] with drift          : -215.5316
##  ARIMA(2,0,0)(0,1,1)[12]                     : Inf
##  ARIMA(2,0,0)(0,1,1)[12] with drift          : Inf
##  ARIMA(2,0,0)(0,1,2)[12]                     : Inf
##  ARIMA(2,0,0)(0,1,2)[12] with drift          : Inf
##  ARIMA(2,0,0)(1,1,0)[12]                     : -231.7311
##  ARIMA(2,0,0)(1,1,0)[12] with drift          : -237.7276
##  ARIMA(2,0,0)(1,1,1)[12]                     : Inf
##  ARIMA(2,0,0)(1,1,1)[12] with drift          : Inf
##  ARIMA(2,0,0)(1,1,2)[12]                     : Inf
##  ARIMA(2,0,0)(1,1,2)[12] with drift          : Inf
##  ARIMA(2,0,0)(2,1,0)[12]                     : -240.8601
##  ARIMA(2,0,0)(2,1,0)[12] with drift          : Inf
##  ARIMA(2,0,0)(2,1,1)[12]                     : Inf
##  ARIMA(2,0,0)(2,1,1)[12] with drift          : Inf
##  ARIMA(2,0,1)(0,1,0)[12]                     : -209.4909
##  ARIMA(2,0,1)(0,1,0)[12] with drift          : Inf
##  ARIMA(2,0,1)(0,1,1)[12]                     : Inf
##  ARIMA(2,0,1)(0,1,1)[12] with drift          : Inf
##  ARIMA(2,0,1)(0,1,2)[12]                     : Inf
##  ARIMA(2,0,1)(0,1,2)[12] with drift          : Inf
##  ARIMA(2,0,1)(1,1,0)[12]                     : -229.5441
##  ARIMA(2,0,1)(1,1,0)[12] with drift          : -235.5931
##  ARIMA(2,0,1)(1,1,1)[12]                     : Inf
##  ARIMA(2,0,1)(1,1,1)[12] with drift          : Inf
##  ARIMA(2,0,1)(2,1,0)[12]                     : Inf
##  ARIMA(2,0,1)(2,1,0)[12] with drift          : -248.3803
##  ARIMA(2,0,2)(0,1,0)[12]                     : -207.5381
##  ARIMA(2,0,2)(0,1,0)[12] with drift          : -212.6273
##  ARIMA(2,0,2)(0,1,1)[12]                     : Inf
##  ARIMA(2,0,2)(0,1,1)[12] with drift          : Inf
```

```
##  ARIMA(2,0,2)(1,1,0)[12]                    : -227.318
##  ARIMA(2,0,2)(1,1,0)[12] with drift         : -234.5348
##  ARIMA(2,0,3)(0,1,0)[12]                    : Inf
##  ARIMA(2,0,3)(0,1,0)[12] with drift         : Inf
##  ARIMA(3,0,0)(0,1,0)[12]                    : -209.5953
##  ARIMA(3,0,0)(0,1,0)[12] with drift         : -214.1661
##  ARIMA(3,0,0)(0,1,1)[12]                    : Inf
##  ARIMA(3,0,0)(0,1,1)[12] with drift         : Inf
##  ARIMA(3,0,0)(0,1,2)[12]                    : Inf
##  ARIMA(3,0,0)(0,1,2)[12] with drift         : Inf
##  ARIMA(3,0,0)(1,1,0)[12]                    : Inf
##  ARIMA(3,0,0)(1,1,0)[12] with drift         : -236.0923
##  ARIMA(3,0,0)(1,1,1)[12]                    : Inf
##  ARIMA(3,0,0)(1,1,1)[12] with drift         : Inf
##  ARIMA(3,0,0)(2,1,0)[12]                    : -238.8435
##  ARIMA(3,0,0)(2,1,0)[12] with drift         : -248.6902
##  ARIMA(3,0,1)(0,1,0)[12]                    : -207.412
##  ARIMA(3,0,1)(0,1,0)[12] with drift         : -212.0937
##  ARIMA(3,0,1)(0,1,1)[12]                    : Inf
##  ARIMA(3,0,1)(0,1,1)[12] with drift         : Inf
##  ARIMA(3,0,1)(1,1,0)[12]                    : Inf
##  ARIMA(3,0,1)(1,1,0)[12] with drift         : -233.946
##  ARIMA(3,0,2)(0,1,0)[12]                    : Inf
##  ARIMA(3,0,2)(0,1,0)[12] with drift         : Inf
##  ARIMA(4,0,0)(0,1,0)[12]                    : -207.4423
##  ARIMA(4,0,0)(0,1,0)[12] with drift         : -212.4944
##  ARIMA(4,0,0)(0,1,1)[12]                    : Inf
##  ARIMA(4,0,0)(0,1,1)[12] with drift         : Inf
##  ARIMA(4,0,0)(1,1,0)[12]                    : -227.3253
##  ARIMA(4,0,0)(1,1,0)[12] with drift         : -234.386
##  ARIMA(4,0,1)(0,1,0)[12]                    : Inf
##  ARIMA(4,0,1)(0,1,0)[12] with drift         : Inf
##  ARIMA(5,0,0)(0,1,0)[12]                    : -207.8251
##  ARIMA(5,0,0)(0,1,0)[12] with drift         : -211.4607
##
##
##
##  Best model: ARIMA(1,0,0)(2,1,0)[12] with drift
```

Aqui, é usada a função *auto.arima*, presente no pacote "forecast". Com ela, são testados diversos modelos, e é selecionado o melhor modelo candidato o com significância estatística e menor Critério de Informação de Akaike Corrigido (AICc). O modelo escolhido é ARIMA(1,0,0)(2,1,0)[12], com drift. Ou seja, ordem 1 na parte autoregressiva, 0 graus de diferenciação envolvido e ordem 0 da parte média móvel. Na parte sazonal, temos ordem 2 na parte autoregressiva, 1 grau de diferenciação envolvido e ordem 0 na parte de média móvel. Entretanto, percebemos que dois modelos tiveram tabém AICc menor em comparação com outros modelos: ARIMA(1,0,1)(2,1,0)[12] com drift e ARIMA(0,0,3)(2,1,0)[12] com drift. Portanto, decidimos testá-los também. O argumento

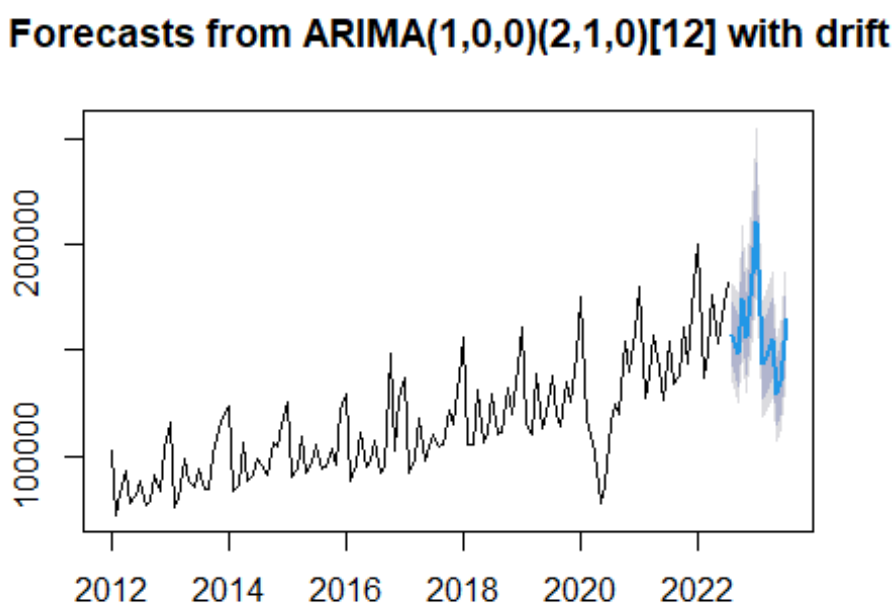
lambda = 0 indica que há transformação Box-Cox nos dados, que toma o log dos valores e depois faz a situação inversa (exponencial) para entregar o resultado das previsões.

```
mod_arrecad1 <- Arima(arrecad_ts, order=c(0,0,3),
                      seasonal=c(2,1,0),include.drift=T,
                      lambda=0)

mod_arrecad2 <- Arima(arrecad_ts, order=c(1,0,1),
                      seasonal=c(2,1,0),include.drift=T,
                      lambda=0)

list(mod_arrecad)

## [[1]]
## Series: arrecad_ts
## ARIMA(1,0,0)(2,1,0)[12] with drift
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ar1     sar1     sar2   drift
##       0.6364  -0.6382  -0.4071  0.0049
## s.e.  0.0758   0.0914   0.0986  0.0008
##
## sigma^2 = 0.005776:  log likelihood = 131.54
## AIC=-253.07   AICc=-252.52   BIC=-239.35

list(mod_arrecad1)

## [[1]]
## Series: arrecad_ts
## ARIMA(0,0,3)(2,1,0)[12] with drift
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ma1     ma2     ma3     sar1     sar2   drift
##       0.5769  0.4587  0.3317  -0.6474  -0.4112  0.0049
## s.e.  0.0893  0.0943  0.0985   0.0934   0.1008  0.0007
##
## sigma^2 = 0.005736:  log likelihood = 132.8
## AIC=-251.59   AICc=-250.55   BIC=-232.38

list(mod_arrecad2)

## [[1]]
## Series: arrecad_ts
## ARIMA(1,0,1)(2,1,0)[12] with drift
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ar1      ma1     sar1     sar2   drift
##       0.6755  -0.0632  -0.6432  -0.4098  0.0049
```

```
## s.e.  0.1090   0.1374   0.0920   0.0989  0.0009
##
## sigma^2 = 0.005813:  log likelihood = 131.64
## AIC=-251.28   AICc=-250.5   BIC=-234.81
```

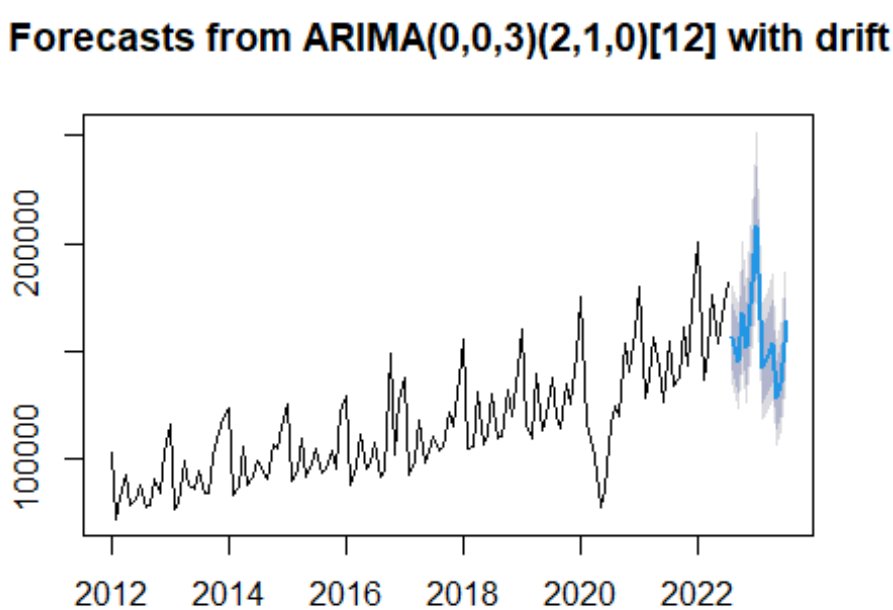
Previsão

```
previsao <- forecast(mod_arrecad, h=12, level=95)
previsao

##          Point Forecast    Lo 95    Hi 95
## Aug 2022       156959.6 135236.7 182171.9
## Sep 2022       148300.5 124297.1 176939.2
## Oct 2022       174320.3 144648.7 210078.2
## Nov 2022       156196.9 129104.7 188974.3
## Dec 2022       185482.9 153072.1 224756.1
## Jan 2023       210974.8 174000.3 255806.2
## Feb 2023       143283.9 118142.7 173775.3
## Mar 2023       146721.2 120964.4 177962.4
## Apr 2023       154877.8 127683.8 187863.5
## May 2023       128983.0 106333.9 156456.3
## Jun 2023       135190.9 111451.0 163987.6
## Jul 2023       164600.3 135695.6 199661.9

plot(forecast(mod_arrecad, h=12))
```




```
previsao1 <- forecast(mod_arrecad1, h=24, level=95)
previsao1
```

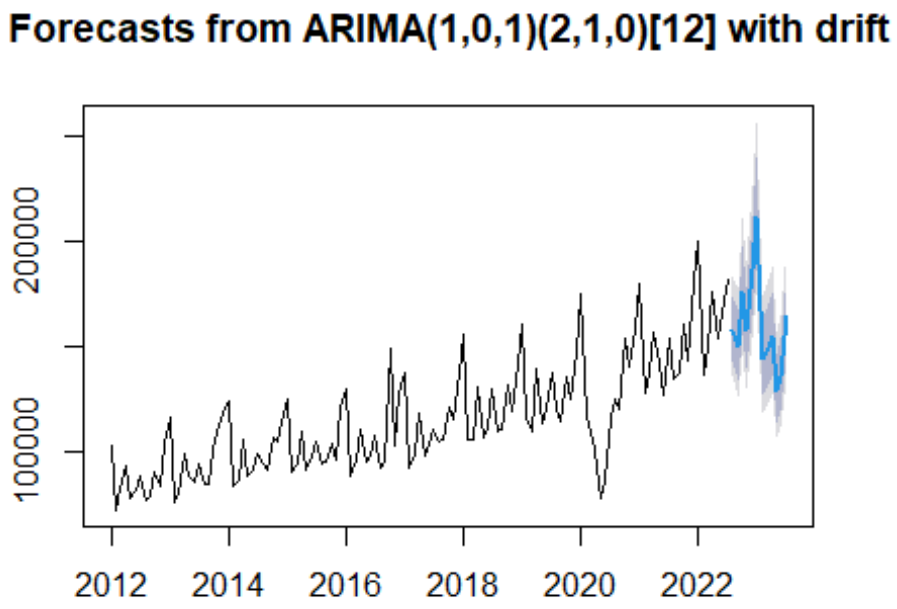
```
##          Point Forecast    Lo 95    Hi 95
## Aug 2022       156069.4 134539.7 181044.5
## Sep 2022       145294.3 122411.2 172455.1
## Oct 2022       167324.0 139146.6 201207.5
## Nov 2022       152063.8 125641.8 184042.4
## Dec 2022       182104.3 150462.5 220400.3
## Jan 2023       208444.5 172225.9 252279.7
## Feb 2023       142109.1 117416.7 171994.2
## Mar 2023       145767.4 120439.4 176421.8
## Apr 2023       153923.5 127178.3 186293.1
## May 2023       128271.5 105983.5 155246.6
## Jun 2023       134406.5 111052.5 162671.8
## Jul 2023       163941.4 135455.5 198417.7
## Aug 2023       154759.8 126971.3 188630.1
## Sep 2023       149425.2 122314.0 182545.6
## Oct 2023       180691.3 147695.3 221058.9
## Nov 2023       163515.8 133556.4 200195.6
## Dec 2023       194048.7 158495.2 237577.7
## Jan 2024       219369.2 179176.4 268578.0
## Feb 2024       152055.2 124195.7 186164.2
## Mar 2024       161740.5 132106.4 198022.1
## Apr 2024       180641.2 147544.1 221162.5
## May 2024       157410.4 128569.7 192720.7
## Jun 2024       155698.1 127171.1 190624.2
## Jul 2024       184754.6 150903.9 226198.7

plot(forecast(mod_arrecad1, h=12))
```



```r
previsao2 <- forecast(mod_arrecad2, h=24, level=95)
previsao2
```

```
##          Point Forecast    Lo 95    Hi 95
## Aug 2022       158138.8 136188.2 183627.4
## Sep 2022       149603.4 125558.4 178253.2
## Oct 2022       175840.7 146025.7 211743.1
## Nov 2022       157409.3 130115.5 190428.4
## Dec 2022       186594.0 153919.9 226204.2
## Jan 2023       212025.6 174733.9 257276.1
## Feb 2023       143844.2 118493.8 174618.1
## Mar 2023       147102.0 121153.9 178607.6
## Apr 2023       155058.0 127695.2 188284.3
## May 2023       129032.9 106258.3 156688.7
## Jun 2023       135138.1 111283.9 164105.5
## Jul 2023       164623.8 135563.8 199913.3
## Aug 2023       156010.7 127509.5 190882.6
## Sep 2023       151403.6 123403.3 185757.2
## Oct 2023       184347.6 150067.6 226458.2
## Nov 2023       165914.0 134985.4 203929.2
## Dec 2023       196145.8 159540.4 241150.0
## Jan 2024       221171.8 179874.9 271950.0
## Feb 2024       153021.1 124442.5 188162.8
## Mar 2024       162546.9 132186.0 199881.2
## Apr 2024       181349.9 147475.3 223005.4
## May 2024       157908.2 128411.7 194180.2
## Jun 2024       156180.6 127006.5 192056.2
## Jul 2024       185300.5 150686.7 227865.3
```

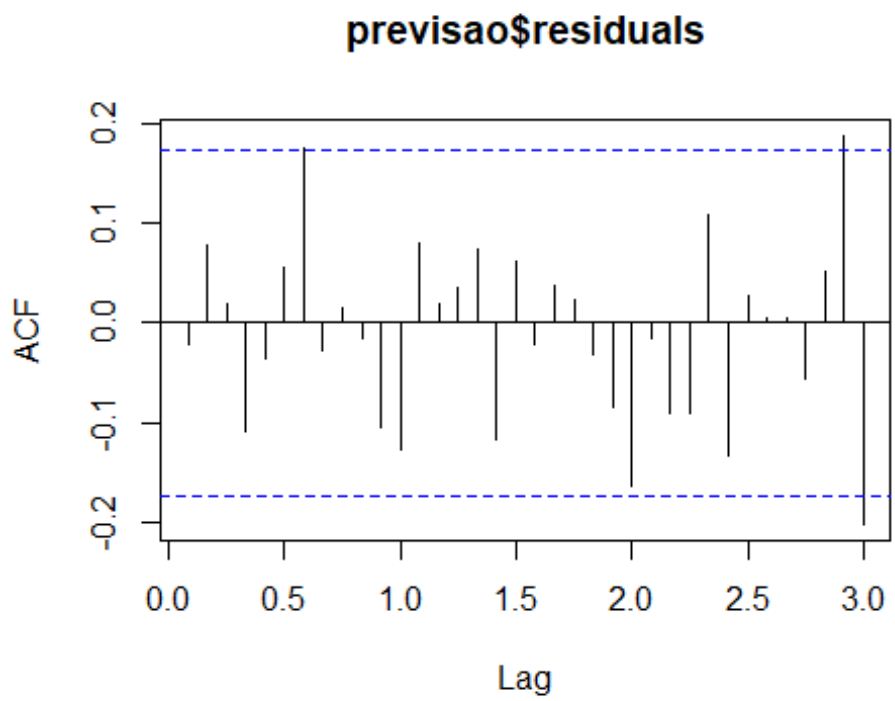
```r
plot(forecast(mod_arrecad2, h=12))
```



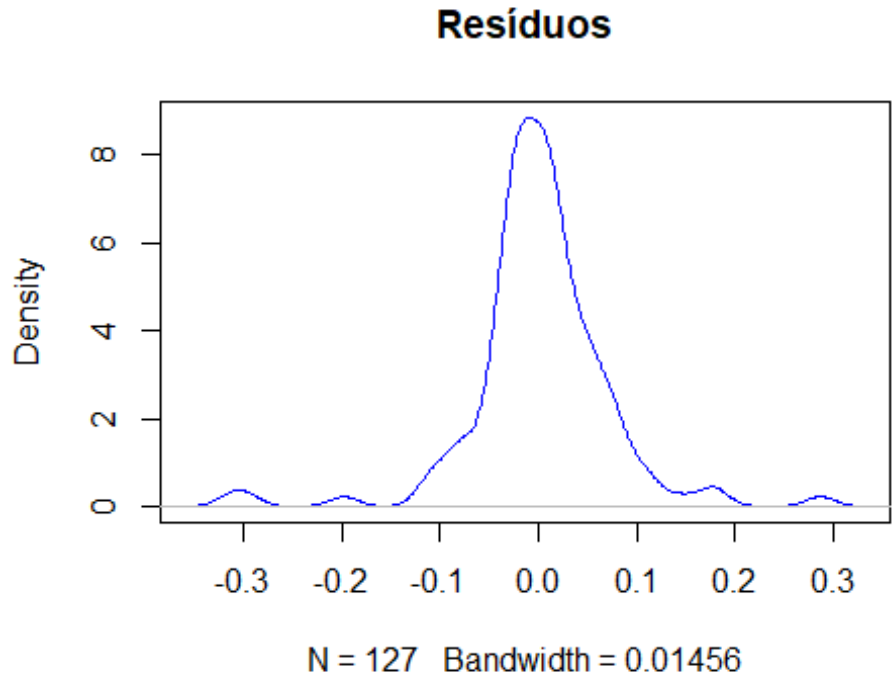
Usamos a função *forecast* para fazer a previsão da série temporal usando os modelos candidatos obtidos pela função *auto.arima*. Previmos 12 passos à frente e com um intervalo de confiança de 95%.

Testes do Modelo 1 - Escolhido pelo auto.arima

```
# Testes de autocorrelação dos resíduos
Acf(previsao$residuals, lag.max=36)
```

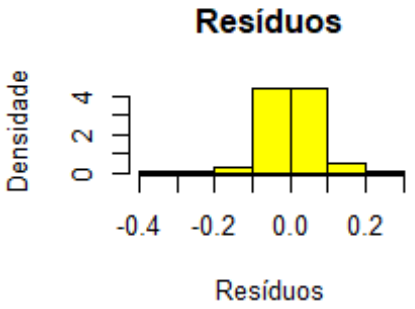


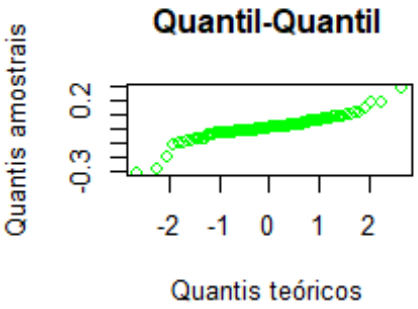

```
plot(density(previsao$residuals,
             kernel = c("gaussian")),
     main="Resíduos", col="blue")
```



```r
Box.test(previsao$residuals,lag=8,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao$residuals
## X-squared = 7.3494, df = 3, p-value = 0.06156

Box.test(previsao$residuals,lag=10,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao$residuals
## X-squared = 7.4134, df = 5, p-value = 0.1917

Box.test(previsao$residuals,lag=12,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao$residuals
## X-squared = 11.221, df = 7, p-value = 0.1293

# Teste de heterocedasticidade
ArchTest(previsao$residuals)

##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  previsao$residuals
## Chi-squared = 22.178, df = 12, p-value = 0.03558

# Teste e análise da normalidade
shapiro.test(previsao$residuals)

##
##  Shapiro-Wilk normality test
##
## data:  previsao$residuals
## W = 0.87745, p-value = 7.889e-09

## Análise visual
par(mfrow=c(2,2))
hist(previsao$residuals, freq=F,
ylab='Densidade', xlab='Resíduos', main='Resíduos', col="yellow")

## Função de densidade estimada
qqnorm(previsao$residuals,
```

```
      ylab='Quantis amostrais', xlab='Quantis teóricos', main='Quantil-
Quantil', col="green")
```



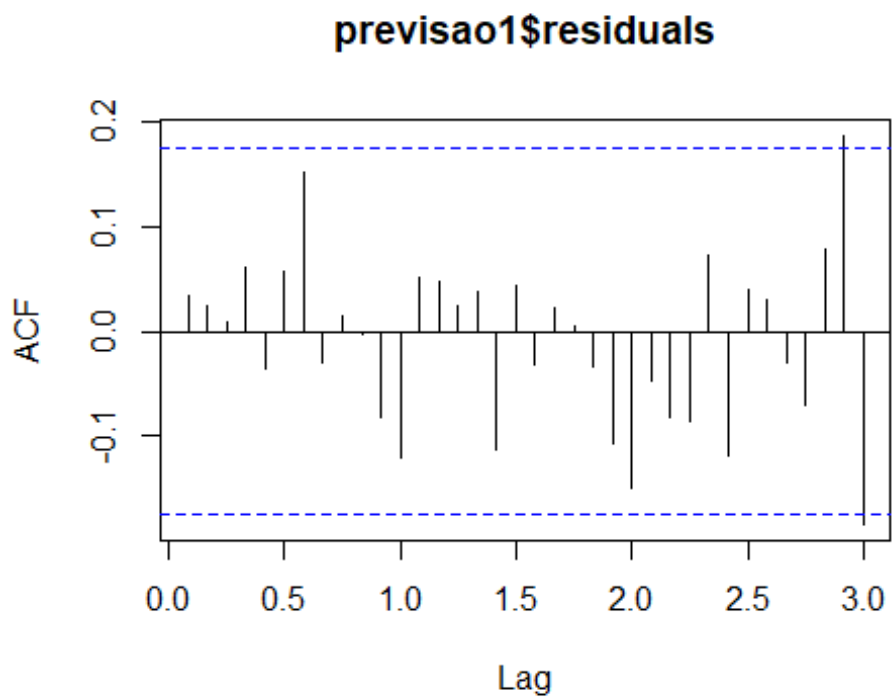


O teste de autocorrelação busca entender se a autocorrelação é estatisticamente nula, ou seja, se os resíduos do modelo são aproximadamente ruído branco. Testamos a heterocedasticidade com o teste ARCH, que busca entender se os resíduos são homocedásticos. Avaliamos se a distribuição dos resíduos seguem aproximadamente uma distribuição normal. Por fim, com a análise visual e com a função de densidade estimada, podemos ver se há normalidade nos resíduos.

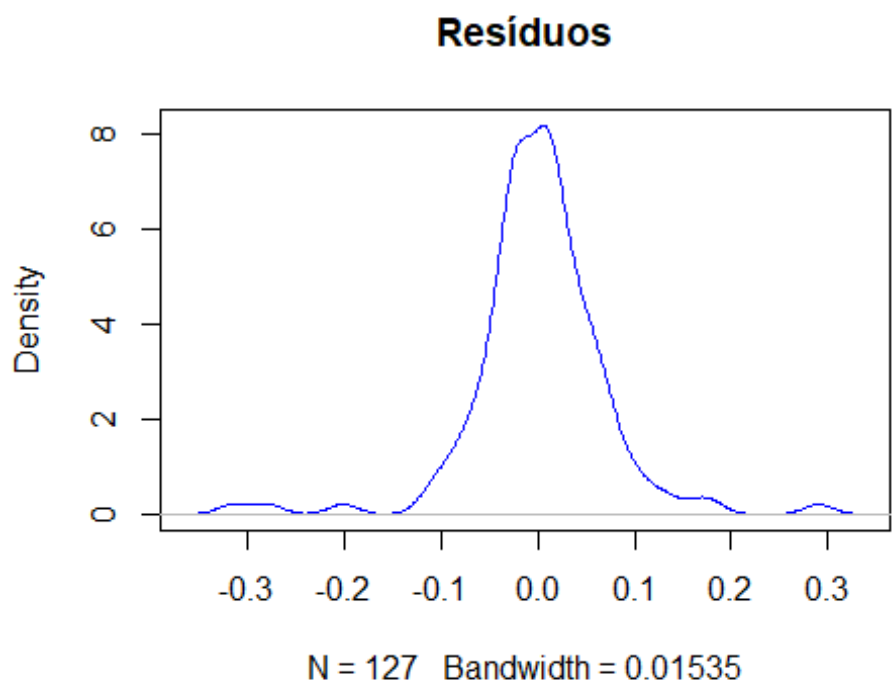
No teste de Ljung-Box, observamos que os valores-p para os diferentes lags (8, 10, 12) não são estatisticamente significativos (todos acima de 0.05). Isso sugere que não há evidência suficiente para rejeitar a hipótese nula de independência dos resíduos, ou seja, eles se aproximam de ruído branco. No teste de heterocedasticidade, o valor-p é 0.03558, e portanto há efeitos de heterocedasticidade nos resíduos. No teste de normalidade de Shapiro-Wilk, o valor-p é extremamente baixo (7.889e-09), indicando que os resíduos não seguem uma distribuição normal. Ou seja, os resíduos não são normalmente distribuídos.. Essas análises indicam que o modelo selecionado pode não ser totalmente adequado, pois os resíduos apresentam violações em relação à heterocedasticidade e normalidade. Talvez seja necessário considerar outras alternativas de modelagem ou explorar técnicas adicionais para melhorar a adequação do modelo aos dados.

Testes do Modelo 2

```
# Testes de autocorrelação dos resíduos
Acf(previsao1$residuals, lag.max=36)
```

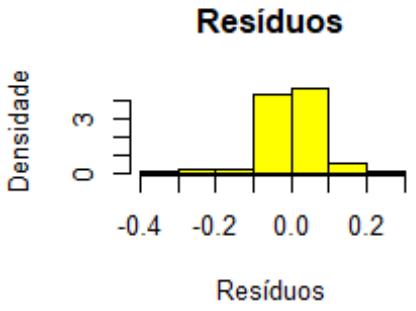


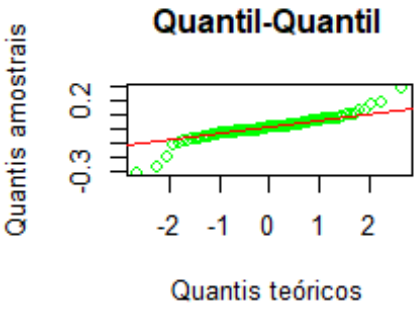

```
plot(density(previsao1$residuals,
             kernel = c("gaussian")),
     main="Resíduos", col="blue")
```



```r
Box.test(previsao1$residuals,lag=8,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao1$residuals
## X-squared = 4.6057, df = 3, p-value = 0.2031

Box.test(previsao1$residuals,lag=10,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao1$residuals
## X-squared = 4.6325, df = 5, p-value = 0.4623

Box.test(previsao1$residuals,lag=12,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao1$residuals
## X-squared = 7.6352, df = 7, p-value = 0.3659

# Teste de heterocedasticidade
ArchTest(previsao1$residuals)

##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  previsao1$residuals
## Chi-squared = 20.346, df = 12, p-value = 0.06082

# Teste e análise da normalidade
shapiro.test(previsao1$residuals)

##
##  Shapiro-Wilk normality test
##
## data:  previsao1$residuals
## W = 0.88554, p-value = 1.905e-08

# Análise visual
par(mfrow=c(2,2))
hist(previsao1$residuals, freq=F,
ylab='Densidade', xlab='Resíduos', main='Resíduos', col="yellow")

# Função de densidade estimada
qqnorm(previsao1$residuals,
```

```
       ylab='Quantis amostrais', xlab='Quantis teóricos', main='Quantil-
Quantil', col="green")

qqline(previsao1$residuals, col="red")
```

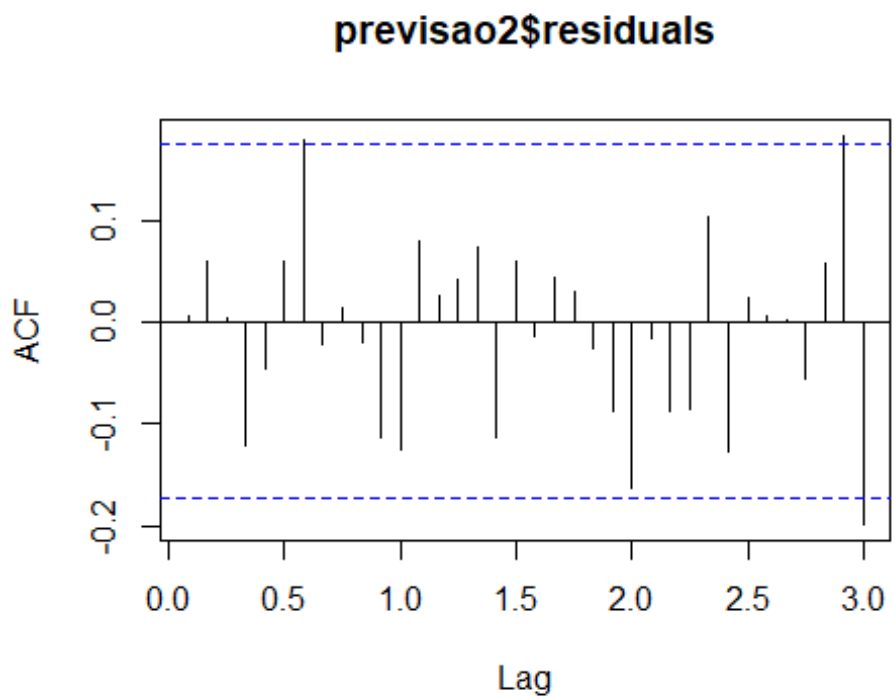




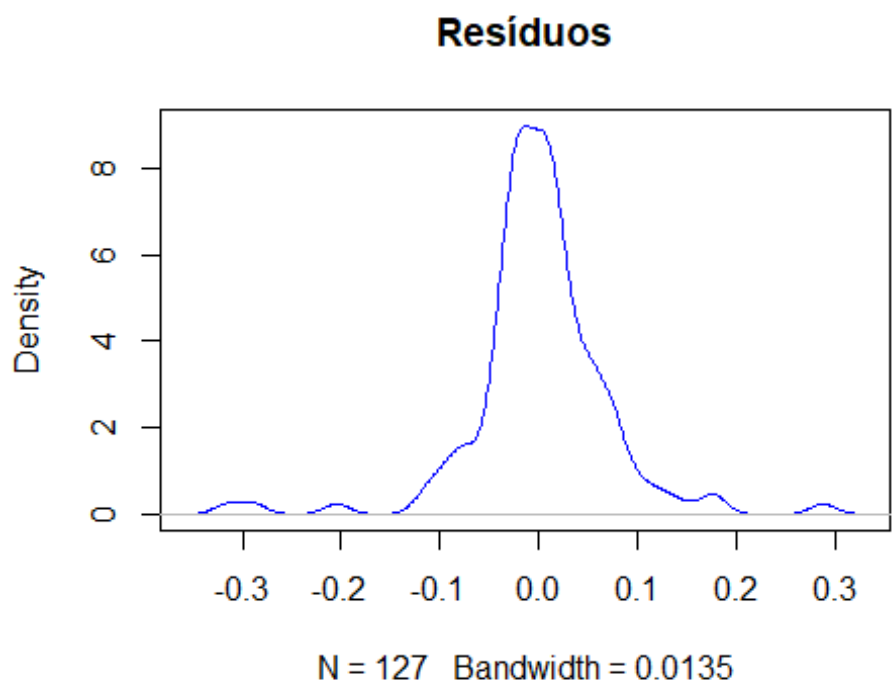
No teste de Ljung-Box, observamos que os valores-p para os diferentes lags (8, 10, 12) se situam acima de 0.05. Isso sugere que há independência dos resíduos. No teste de heterocedasticidade, o valor-p é 0.06082, que está próximo de 0.05. Assim, os resíduos podem ser considerados homocedásticos. No teste de normalidade de Shapiro-Wilk, o valor-p é extremamente baixo (0.00000001905), indicando que os resíduos não seguem uma distribuição normal.

Testes do Modelo 3

```
# Testes de autocorrelação dos resíduos
Acf(previsao2$residuals, lag.max=36)
```

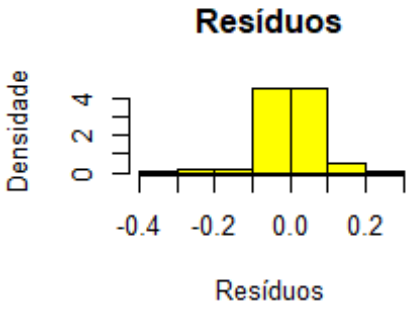


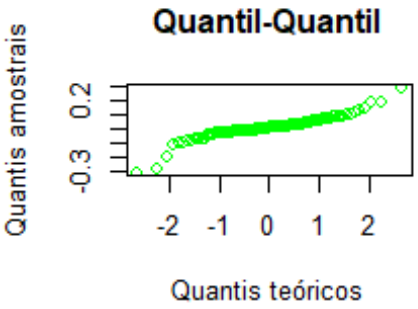

```
plot(density(previsao2$residuals,
             kernel = c("gaussian")),
     main="Resíduos", col="blue")
```



```r
Box.test(previsao2$residuals,lag=8,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao2$residuals
## X-squared = 7.6449, df = 3, p-value = 0.05395

Box.test(previsao2$residuals,lag=10,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao2$residuals
## X-squared = 7.7204, df = 5, p-value = 0.1723

Box.test(previsao2$residuals,lag=12,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  previsao2$residuals
## X-squared = 11.764, df = 7, p-value = 0.1086

# Teste de heterocedasticidade
ArchTest(previsao2$residuals)

##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  previsao2$residuals
## Chi-squared = 22.609, df = 12, p-value = 0.03123

# Teste e análise da normalidade
shapiro.test(previsao2$residuals)

##
##  Shapiro-Wilk normality test
##
## data:  previsao2$residuals
## W = 0.87594, p-value = 6.721e-09

# Análise visual
par(mfrow=c(2,2))
hist(previsao2$residuals, freq=F,
ylab='Densidade', xlab='Resíduos', main='Resíduos', col="yellow")

# Função de densidade estimada
qqnorm(previsao2$residuals,
```

```
      ylab='Quantis amostrais', xlab='Quantis teóricos', main='Quantil-Quantil', col="green")
```





Mais uma vez, os resíduos passam no teste de Ljung-Box, porém não passam no teste de homocedasticidade, sugerindo efeitos de heterocedasticidade. No teste de normalidade de Shapiro-Wilk, o valor-p é extremamente baixo (0.000000006721), indicando que os resíduos não seguem uma distribuição normal.

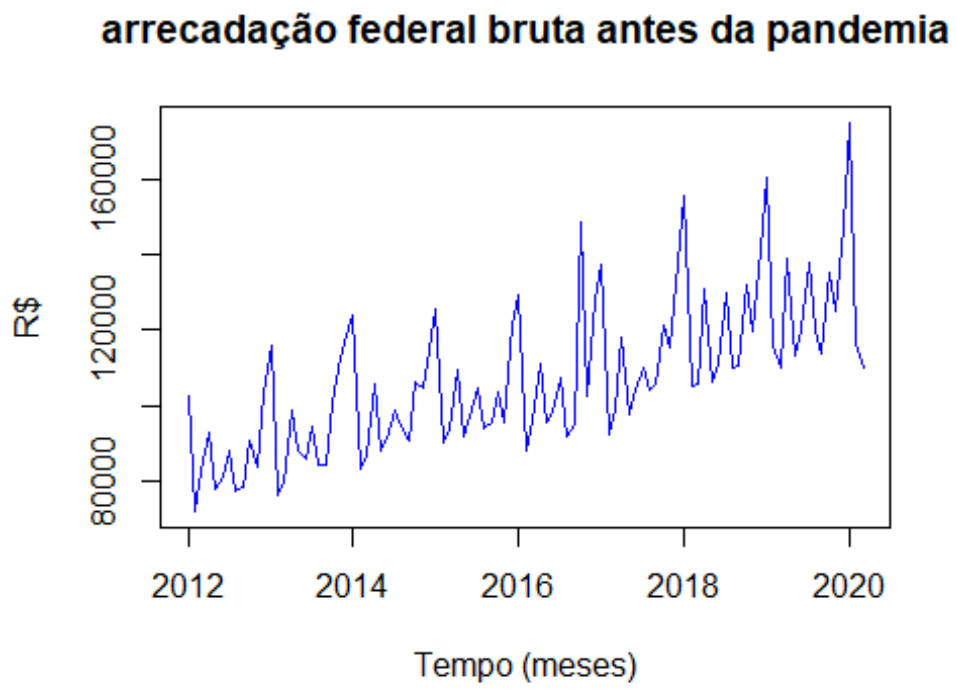
Após a análise de resíduos, concluímos que tanto o modelo selecionado automaticamente pelo auto.arima quanto os outros dois modelos escolhidos manualmente possuem erros, em especial no que tange à normalidade da distribuição dos resíduos. Já em relação ao teste de homocedasticidade, o modelo selecionado automaticamente (mod_arrecad) e o mod_arrecad2 falharam, com p-valor < 5%, enquanto que o ARIMA(0,0,3)(2,1,0) (=mod_arrecad1) passou. Assim, o mod_arrecad1, que é um ARIMA(0,0,3)(2,1,0) parece ser o melhor, tanto em termos de significância dos coeficientes quanto na análise dos resíduos. Apesar de não ser ideal, pois possui distribuição não normal, demonstrou ser o melhor dos modelos analisados. Portanto, ele será o escolhido para a Análise de Intervenção.

2ª Etapa: Teste do Efeito Covid-19

Tendo feito a previsão da série sem considerar os efeitos da pandemia, agora vamos analisar o efeito do isolamento social decorrente da Covid-19 na série. Para isso, seguiremos o seguintes passos: (OBS: prosseguiremos diretamente para a seleção da série pré-intervenção)

Restringindo a amostra ao período pré-intervenção e plotando a amostra

```
# Definindo a amostra
arrecad_pre <- ts(arrecadacao[2:100,3],
                      start=c(2012,1),frequency=12)

# Conhecendo o tamanho da amostra pré
length(arrecad_pre)

## [1] 99

# Plotando
plot.ts(arrecad_pre, xlab="Tempo (meses)",
        ylab='R$',
        main="arrecadação federal bruta antes da pandemia",
        col="blue",
        type="l")
```



Aqui, selecionamos o período do primeiro trimestre de 2012 até o quarto trimestre de 2019, que corresponde ao período antes de começar o isolamento social graças à pandemia de Covid-19. O isolamento começou gradualmente em algumas instituições entre os dias 12 e 20 de março.

Aplicando a seleção automática

```
mod_pre <- auto.arima(arrecad_pre, lambda=0,
                        stepwise=FALSE, trace=TRUE,
```

```
                        approximation=FALSE, allowdrift=TRUE,
                        allowmean=TRUE, ic=c("aicc"))

##
##  ARIMA(0,0,0)(0,1,0)[12]                    : -174.1984
##  ARIMA(0,0,0)(0,1,0)[12] with drift         : -218.1487
##  ARIMA(0,0,0)(0,1,1)[12]                    : -172.8953
##  ARIMA(0,0,0)(0,1,1)[12] with drift         : Inf
##  ARIMA(0,0,0)(0,1,2)[12]                    : -172.8107
##  ARIMA(0,0,0)(0,1,2)[12] with drift         : Inf
##  ARIMA(0,0,0)(1,1,0)[12]                    : -173.2233
##  ARIMA(0,0,0)(1,1,0)[12] with drift         : -230.927
##  ARIMA(0,0,0)(1,1,1)[12]                    : Inf
##  ARIMA(0,0,0)(1,1,1)[12] with drift         : Inf
##  ARIMA(0,0,0)(1,1,2)[12]                    : Inf
##  ARIMA(0,0,0)(1,1,2)[12] with drift         : Inf
##  ARIMA(0,0,0)(2,1,0)[12]                    : -175.0263
##  ARIMA(0,0,0)(2,1,0)[12] with drift         : -239.7366
##  ARIMA(0,0,0)(2,1,1)[12]                    : Inf
##  ARIMA(0,0,0)(2,1,1)[12] with drift         : -247.5615
##  ARIMA(0,0,0)(2,1,2)[12]                    : Inf
##  ARIMA(0,0,0)(2,1,2)[12] with drift         : Inf
##  ARIMA(0,0,1)(0,1,0)[12]                    : -188.1018
##  ARIMA(0,0,1)(0,1,0)[12] with drift         : -217.3493
##  ARIMA(0,0,1)(0,1,1)[12]                    : -186.8578
##  ARIMA(0,0,1)(0,1,1)[12] with drift         : Inf
##  ARIMA(0,0,1)(0,1,2)[12]                    : -186.244
##  ARIMA(0,0,1)(0,1,2)[12] with drift         : Inf
##  ARIMA(0,0,1)(1,1,0)[12]                    : -187.1703
##  ARIMA(0,0,1)(1,1,0)[12] with drift         : -231.0383
##  ARIMA(0,0,1)(1,1,1)[12]                    : -185.4835
##  ARIMA(0,0,1)(1,1,1)[12] with drift         : Inf
##  ARIMA(0,0,1)(1,1,2)[12]                    : Inf
##  ARIMA(0,0,1)(1,1,2)[12] with drift         : Inf
##  ARIMA(0,0,1)(2,1,0)[12]                    : -186.0859
##  ARIMA(0,0,1)(2,1,0)[12] with drift         : -238.2782
##  ARIMA(0,0,1)(2,1,1)[12]                    : Inf
##  ARIMA(0,0,1)(2,1,1)[12] with drift         : -245.8864
##  ARIMA(0,0,1)(2,1,2)[12]                    : Inf
##  ARIMA(0,0,1)(2,1,2)[12] with drift         : Inf
##  ARIMA(0,0,2)(0,1,0)[12]                    : -188.341
##  ARIMA(0,0,2)(0,1,0)[12] with drift         : -215.4129
##  ARIMA(0,0,2)(0,1,1)[12]                    : -188.3871
##  ARIMA(0,0,2)(0,1,1)[12] with drift         : Inf
##  ARIMA(0,0,2)(0,1,2)[12]                    : -187.1665
##  ARIMA(0,0,2)(0,1,2)[12] with drift         : Inf
##  ARIMA(0,0,2)(1,1,0)[12]                    : -188.8715
##  ARIMA(0,0,2)(1,1,0)[12] with drift         : -228.7958
##  ARIMA(0,0,2)(1,1,1)[12]                    : -186.7171
##  ARIMA(0,0,2)(1,1,1)[12] with drift         : Inf
```

```
##  ARIMA(0,0,2)(1,1,2)[12]                     : Inf
##  ARIMA(0,0,2)(1,1,2)[12] with drift          : Inf
##  ARIMA(0,0,2)(2,1,0)[12]                     : -186.7997
##  ARIMA(0,0,2)(2,1,0)[12] with drift          : -235.9853
##  ARIMA(0,0,2)(2,1,1)[12]                     : Inf
##  ARIMA(0,0,2)(2,1,1)[12] with drift          : -243.8443
##  ARIMA(0,0,3)(0,1,0)[12]                     : -192.0771
##  ARIMA(0,0,3)(0,1,0)[12] with drift          : -214.0649
##  ARIMA(0,0,3)(0,1,1)[12]                     : -195.8947
##  ARIMA(0,0,3)(0,1,1)[12] with drift          : Inf
##  ARIMA(0,0,3)(0,1,2)[12]                     : -195.2035
##  ARIMA(0,0,3)(0,1,2)[12] with drift          : Inf
##  ARIMA(0,0,3)(1,1,0)[12]                     : -196.9684
##  ARIMA(0,0,3)(1,1,0)[12] with drift          : -228.9839
##  ARIMA(0,0,3)(1,1,1)[12]                     : -194.6736
##  ARIMA(0,0,3)(1,1,1)[12] with drift          : Inf
##  ARIMA(0,0,3)(2,1,0)[12]                     : -194.6787
##  ARIMA(0,0,3)(2,1,0)[12] with drift          : -235.3067
##  ARIMA(0,0,4)(0,1,0)[12]                     : -191.2157
##  ARIMA(0,0,4)(0,1,0)[12] with drift          : -211.7556
##  ARIMA(0,0,4)(0,1,1)[12]                     : -194.8525
##  ARIMA(0,0,4)(0,1,1)[12] with drift          : Inf
##  ARIMA(0,0,4)(1,1,0)[12]                     : -195.6469
##  ARIMA(0,0,4)(1,1,0)[12] with drift          : -226.6432
##  ARIMA(0,0,5)(0,1,0)[12]                     : -192.8669
##  ARIMA(0,0,5)(0,1,0)[12] with drift          : -210.0585
##  ARIMA(1,0,0)(0,1,0)[12]                     : -194.6234
##  ARIMA(1,0,0)(0,1,0)[12] with drift          : -217.2183
##  ARIMA(1,0,0)(0,1,1)[12]                     : -205.235
##  ARIMA(1,0,0)(0,1,1)[12] with drift          : Inf
##  ARIMA(1,0,0)(0,1,2)[12]                     : -203.0369
##  ARIMA(1,0,0)(0,1,2)[12] with drift          : Inf
##  ARIMA(1,0,0)(1,1,0)[12]                     : -201.7059
##  ARIMA(1,0,0)(1,1,0)[12] with drift          : -231.0629
##  ARIMA(1,0,0)(1,1,1)[12]                     : -203.0368
##  ARIMA(1,0,0)(1,1,1)[12] with drift          : Inf
##  ARIMA(1,0,0)(1,1,2)[12]                     : Inf
##  ARIMA(1,0,0)(1,1,2)[12] with drift          : Inf
##  ARIMA(1,0,0)(2,1,0)[12]                     : -201.2462
##  ARIMA(1,0,0)(2,1,0)[12] with drift          : Inf
##  ARIMA(1,0,0)(2,1,1)[12]                     : -200.8205
##  ARIMA(1,0,0)(2,1,1)[12] with drift          : -245.8097
##  ARIMA(1,0,0)(2,1,2)[12]                     : Inf
##  ARIMA(1,0,0)(2,1,2)[12] with drift          : Inf
##  ARIMA(1,0,1)(0,1,0)[12]                     : Inf
##  ARIMA(1,0,1)(0,1,0)[12] with drift          : -218.0334
##  ARIMA(1,0,1)(0,1,1)[12]                     : Inf
##  ARIMA(1,0,1)(0,1,1)[12] with drift          : Inf
##  ARIMA(1,0,1)(0,1,2)[12]                     : Inf
##  ARIMA(1,0,1)(0,1,2)[12] with drift          : Inf
```

```
##  ARIMA(1,0,1)(1,1,0)[12]                     : Inf
##  ARIMA(1,0,1)(1,1,0)[12] with drift          : -230.1632
##  ARIMA(1,0,1)(1,1,1)[12]                     : Inf
##  ARIMA(1,0,1)(1,1,1)[12] with drift          : Inf
##  ARIMA(1,0,1)(1,1,2)[12]                     : Inf
##  ARIMA(1,0,1)(1,1,2)[12] with drift          : Inf
##  ARIMA(1,0,1)(2,1,0)[12]                     : Inf
##  ARIMA(1,0,1)(2,1,0)[12] with drift          : Inf
##  ARIMA(1,0,1)(2,1,1)[12]                     : Inf
##  ARIMA(1,0,1)(2,1,1)[12] with drift          : -244.0485
##  ARIMA(1,0,2)(0,1,0)[12]                     : Inf
##  ARIMA(1,0,2)(0,1,0)[12] with drift          : -216.0065
##  ARIMA(1,0,2)(0,1,1)[12]                     : Inf
##  ARIMA(1,0,2)(0,1,1)[12] with drift          : Inf
##  ARIMA(1,0,2)(0,1,2)[12]                     : Inf
##  ARIMA(1,0,2)(0,1,2)[12] with drift          : Inf
##  ARIMA(1,0,2)(1,1,0)[12]                     : Inf
##  ARIMA(1,0,2)(1,1,0)[12] with drift          : -227.9488
##  ARIMA(1,0,2)(1,1,1)[12]                     : Inf
##  ARIMA(1,0,2)(1,1,1)[12] with drift          : Inf
##  ARIMA(1,0,2)(2,1,0)[12]                     : Inf
##  ARIMA(1,0,2)(2,1,0)[12] with drift          : -234.45
##  ARIMA(1,0,3)(0,1,0)[12]                     : Inf
##  ARIMA(1,0,3)(0,1,0)[12] with drift          : Inf
##  ARIMA(1,0,3)(0,1,1)[12]                     : Inf
##  ARIMA(1,0,3)(0,1,1)[12] with drift          : Inf
##  ARIMA(1,0,3)(1,1,0)[12]                     : Inf
##  ARIMA(1,0,3)(1,1,0)[12] with drift          : -227.1523
##  ARIMA(1,0,4)(0,1,0)[12]                     : Inf
##  ARIMA(1,0,4)(0,1,0)[12] with drift          : Inf
##  ARIMA(2,0,0)(0,1,0)[12]                     : -196.4611
##  ARIMA(2,0,0)(0,1,0)[12] with drift          : -215.1762
##  ARIMA(2,0,0)(0,1,1)[12]                     : -216.9497
##  ARIMA(2,0,0)(0,1,1)[12] with drift          : Inf
##  ARIMA(2,0,0)(0,1,2)[12]                     : -214.7141
##  ARIMA(2,0,0)(0,1,2)[12] with drift          : Inf
##  ARIMA(2,0,0)(1,1,0)[12]                     : -208.2717
##  ARIMA(2,0,0)(1,1,0)[12] with drift          : -228.8463
##  ARIMA(2,0,0)(1,1,1)[12]                     : -214.7117
##  ARIMA(2,0,0)(1,1,1)[12] with drift          : Inf
##  ARIMA(2,0,0)(1,1,2)[12]                     : Inf
##  ARIMA(2,0,0)(1,1,2)[12] with drift          : Inf
##  ARIMA(2,0,0)(2,1,0)[12]                     : -210.7607
##  ARIMA(2,0,0)(2,1,0)[12] with drift          : Inf
##  ARIMA(2,0,0)(2,1,1)[12]                     : -212.5594
##  ARIMA(2,0,0)(2,1,1)[12] with drift          : -243.6841
##  ARIMA(2,0,1)(0,1,0)[12]                     : Inf
##  ARIMA(2,0,1)(0,1,0)[12] with drift          : -216.0144
##  ARIMA(2,0,1)(0,1,1)[12]                     : Inf
##  ARIMA(2,0,1)(0,1,1)[12] with drift          : Inf
```

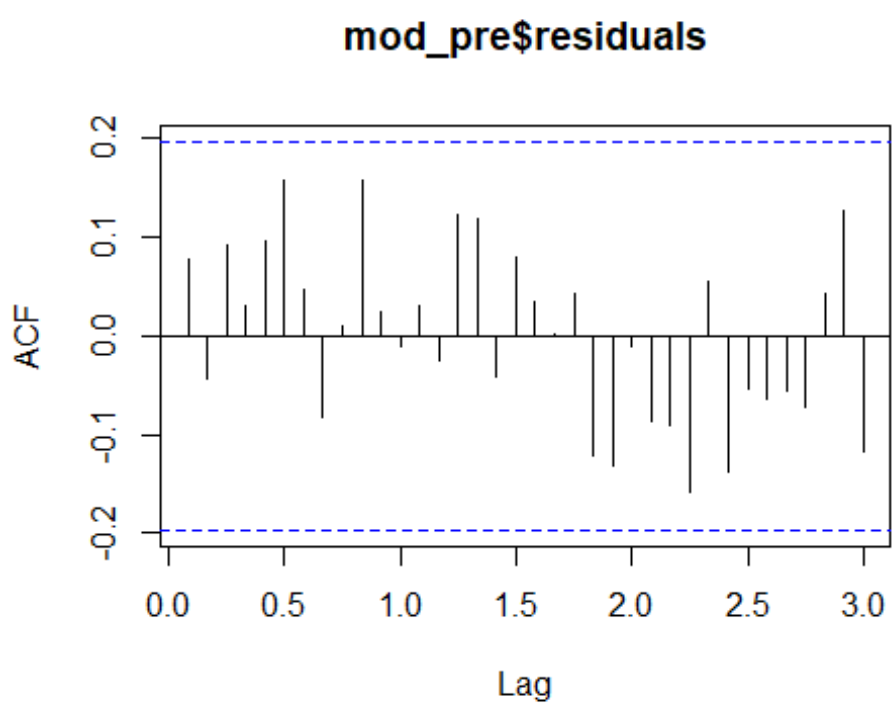
```
##  ARIMA(2,0,1)(0,1,2)[12]                     : Inf
##  ARIMA(2,0,1)(0,1,2)[12] with drift          : Inf
##  ARIMA(2,0,1)(1,1,0)[12]                     : Inf
##  ARIMA(2,0,1)(1,1,0)[12] with drift          : -227.9209
##  ARIMA(2,0,1)(1,1,1)[12]                     : Inf
##  ARIMA(2,0,1)(1,1,1)[12] with drift          : Inf
##  ARIMA(2,0,1)(2,1,0)[12]                     : Inf
##  ARIMA(2,0,1)(2,1,0)[12] with drift          : -233.6024
##  ARIMA(2,0,2)(0,1,0)[12]                     : Inf
##  ARIMA(2,0,2)(0,1,0)[12] with drift          : -214.142
##  ARIMA(2,0,2)(0,1,1)[12]                     : Inf
##  ARIMA(2,0,2)(0,1,1)[12] with drift          : Inf
##  ARIMA(2,0,2)(1,1,0)[12]                     : Inf
##  ARIMA(2,0,2)(1,1,0)[12] with drift          : Inf
##  ARIMA(2,0,3)(0,1,0)[12]                     : Inf
##  ARIMA(2,0,3)(0,1,0)[12] with drift          : Inf
##  ARIMA(3,0,0)(0,1,0)[12]                     : -201.9056
##  ARIMA(3,0,0)(0,1,0)[12] with drift          : -213.9113
##  ARIMA(3,0,0)(0,1,1)[12]                     : Inf
##  ARIMA(3,0,0)(0,1,1)[12] with drift          : Inf
##  ARIMA(3,0,0)(0,1,2)[12]                     : Inf
##  ARIMA(3,0,0)(0,1,2)[12] with drift          : Inf
##  ARIMA(3,0,0)(1,1,0)[12]                     : -217.5821
##  ARIMA(3,0,0)(1,1,0)[12] with drift          : -229.0649
##  ARIMA(3,0,0)(1,1,1)[12]                     : Inf
##  ARIMA(3,0,0)(1,1,1)[12] with drift          : Inf
##  ARIMA(3,0,0)(2,1,0)[12]                     : -221.6136
##  ARIMA(3,0,0)(2,1,0)[12] with drift          : -235.6949
##  ARIMA(3,0,1)(0,1,0)[12]                     : Inf
##  ARIMA(3,0,1)(0,1,0)[12] with drift          : -213.7267
##  ARIMA(3,0,1)(0,1,1)[12]                     : Inf
##  ARIMA(3,0,1)(0,1,1)[12] with drift          : Inf
##  ARIMA(3,0,1)(1,1,0)[12]                     : Inf
##  ARIMA(3,0,1)(1,1,0)[12] with drift          : -227.0569
##  ARIMA(3,0,2)(0,1,0)[12]                     : Inf
##  ARIMA(3,0,2)(0,1,0)[12] with drift          : Inf
##  ARIMA(4,0,0)(0,1,0)[12]                     : -201.2306
##  ARIMA(4,0,0)(0,1,0)[12] with drift          : -211.6065
##  ARIMA(4,0,0)(0,1,1)[12]                     : Inf
##  ARIMA(4,0,0)(0,1,1)[12] with drift          : Inf
##  ARIMA(4,0,0)(1,1,0)[12]                     : -216.4852
##  ARIMA(4,0,0)(1,1,0)[12] with drift          : -226.725
##  ARIMA(4,0,1)(0,1,0)[12]                     : Inf
##  ARIMA(4,0,1)(0,1,0)[12] with drift          : Inf
##  ARIMA(5,0,0)(0,1,0)[12]                     : -202.8491
##  ARIMA(5,0,0)(0,1,0)[12] with drift          : -210.1405
##
##
##
##  Best model: ARIMA(0,0,0)(2,1,1)[12] with drift
```

```
list(mod_pre)

## [[1]]
## Series: arrecad_pre
## ARIMA(0,0,0)(2,1,1)[12] with drift
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          sar1     sar2     sma1   drift
##       -0.0205  -0.1899  -0.8347  0.0044
## s.e.   0.1668   0.1384   0.2871  0.0002
##
## sigma^2 = 0.002589:  log likelihood = 129.15
## AIC=-248.3   AICc=-247.56   BIC=-235.97
```

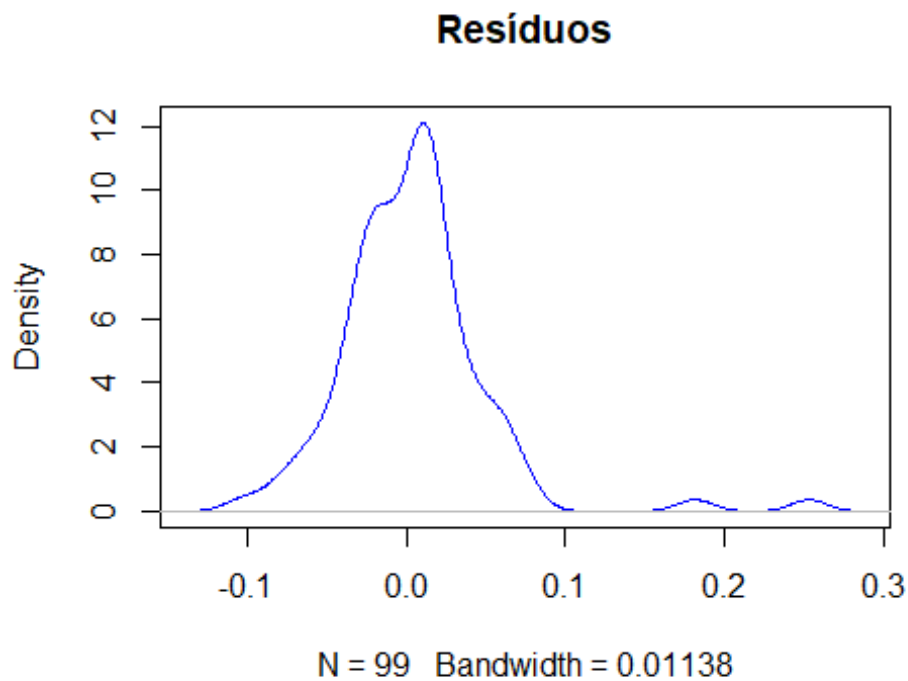
Aplicamos a seleção automática, com a função *auto.arima*, na amostra referente ao período pré-intervenção. Nesse caso, usamos os argumentos stepwise = FALSE (ou seja, não pulamos etapas na seleção do modelo) e approximation = FALSE (não será usada uma aproximação para o cálculo dos coeficientes do modelo ARIMA). Obtivemos um modelo (0,0,0)(2,1,1). Ou seja, Ou seja, ordem 0 na parte autoregressiva, com 0 grau de diferenciação envolvido e ordem 0 da parte média móvel. Na parte sazonal, temos ordem 2 na parte autoregressiva, 1 grau de diferenciação envolvido e ordem 1 na parte de média móvel.

Testes dos resíduos

```
# Testes de autocorrelação dos resíduos
Acf(mod_pre$residuals, lag.max=36)
```

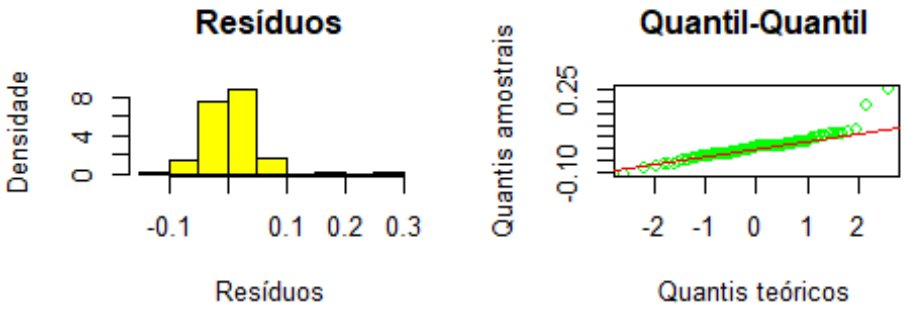


```
plot(density(mod_pre$residuals,
             kernel = c("gaussian")),
     main="Resíduos", col="blue")
```




```
Box.test(mod_pre$residuals,lag=8,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  mod_pre$residuals
## X-squared = 6.4824, df = 3, p-value = 0.09036

Box.test(mod_pre$residuals,lag=10,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  mod_pre$residuals
## X-squared = 9.2743, df = 5, p-value = 0.09861

Box.test(mod_pre$residuals,lag=12,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
```

```
## data:  mod_pre$residuals
## X-squared = 9.3492, df = 7, p-value = 0.2285

# Teste de heterocedasticidade
ArchTest(mod_pre$residuals)

##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  mod_pre$residuals
## Chi-squared = 2.0108, df = 12, p-value = 0.9994

# Teste e análise da normalidade
shapiro.test(mod_pre$residuals)

##
##  Shapiro-Wilk normality test
##
## data:  mod_pre$residuals
## W = 0.8591, p-value = 2.946e-08

# Análise visual
par(mfrow=c(2,2))
hist(mod_pre$residuals, freq=F,
ylab='Densidade', xlab='Resíduos', main='Resíduos', col="yellow")

# Função de densidade estimada
qqnorm(mod_pre$residuals,
       ylab='Quantis amostrais', xlab='Quantis teóricos', main='Quantil-
Quantil', col="green")
qqline(mod_pre$residuals, col="red")
```



O teste de dependência serial indica que os resíduos não mostram evidências significativas de autocorrelação até um lag de 12 períodos, uma vez que os valores-p associados aos testes são maiores que o nível de significância usual de 0,05. Isso significa que o modelo é capaz de capturar a estrutura de dependência temporal dos dados. O teste de heterocedasticidade indica que não há evidências significativas de efeitos de heterocedasticidade nos resíduos do modelo. O valor-p associado ao teste é elevado, acima do nível de significância de 0,05. Portanto, podemos considerar que o modelo é homocedástico. Por fim, o teste de normalidade sugere que os resíduos não seguem uma distribuição normal. O valor-p associado ao teste é menor que 0,05, indicando uma rejeição da hipótese nula de normalidade. Isso pode ser um indício de que o modelo não necessariamente captura completamente todas as características dos dados, especialmente em relação a eventos extremos ou outliers.

Criando a dummy com efeito permanente e temporário

```
# efeito permanente
t <- seq(2, 127)
n <- 127
covid1 <- rep(0,n)
covid1[t>=100] <- 1
covid1

##   [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
##  [38] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
##  [75] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1
```

```
1 1 1
## [112] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0
```

```
# efeitos temporários
covid2 <- rep(0,n)
covid2[105:106] <- 1
covid2
```

```
##   [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
##  [38] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
##  [75] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0
0 0 0
## [112] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
covid3 <- rep(0,n)
covid3[105:113] <- 1
covid3
```

```
##   [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
##  [38] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
##  [75] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1
1 1 1
## [112] 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
covid4 <- rep(0,n)
covid4[105:110] <- 1
covid4
```

```
##   [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
##  [38] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0
##  [75] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1
1 1 0
## [112] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
length(arrecad_ts)
```

```
## [1] 127
```

Aqui, estamos considerando a variável covid2 de março até abril/2020, covid3 de março até dezembro/2020 e covid4 de março até agosto/2020. Pela análise visual, suspeitamos que o impacto da covid tenha sido somente no cenário da variável covid2. De qualquer maneira, iremos testar os outros dois cenários.

Estimando os modelos

```
mod_dummy <- Arima(arrecad_ts, xreg=covid1, order=c(1,0,0),
                     seasonal=c(2,0,1),include.drift=T,
                     lambda=0)
list(mod_dummy)

## [[1]]
## Series: arrecad_ts
## Regression with ARIMA(1,0,0)(2,0,1)[12] errors
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ar1    sar1     sar2     sma1  intercept   drift     xreg
##       0.6352  1.0251  -0.0252  -0.9746    11.3125  0.0051  -0.0426
## s.e.  0.0697  0.1001   0.1001   0.0579     0.1100  0.0005   0.0386
##
## sigma^2 = 0.004357:  log likelihood = 147
## AIC=-278   AICc=-276.78   BIC=-255.25
```

Nem todos os coeficientes são significantes. Portanto testaremos tanto esse mesmo modelo com dummys em outros períodos para o modelo ARIMA(0,0,3)(2,1,0), que foi mais significante e com melhores resíduos na amostra completa, também com dummies em diferentes períodos

```
mod_dummy_cov2 <- Arima(arrecad_ts, xreg=covid2, order=c(1,0,0),
                     seasonal=c(2,0,1),include.drift=T,
                     lambda=0)
list(mod_dummy_cov2)

## [[1]]
## Series: arrecad_ts
## Regression with ARIMA(1,0,0)(2,0,1)[12] errors
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ar1    sar1     sar2     sma1  intercept   drift     xreg
##       0.6390  1.0195  -0.0196  -0.9657    11.3220  0.0048  -0.0152
## s.e.  0.0698  0.0996   0.0996   0.0798     0.1109  0.0004   0.0540
##
## sigma^2 = 0.004431:  log likelihood = 146.42
## AIC=-276.85   AICc=-275.63   BIC=-254.1

mod_dummy_cov3 <- Arima(arrecad_ts, xreg=covid3, order=c(1,0,0),
                     seasonal=c(2,0,1),include.drift=T,
                     lambda=0)
list(mod_dummy_cov3)

## [[1]]
## Series: arrecad_ts
## Regression with ARIMA(1,0,0)(2,0,1)[12] errors
## Box Cox transformation: lambda= 0
```

```
##
## Coefficients:
##          ar1    sar1     sar2     sma1  intercept   drift    xreg
##       0.6388  1.0330  -0.0331  -0.9699    11.3270  0.0047  0.0502
## s.e.  0.0701  0.1008   0.1007   0.0718     0.1109  0.0004  0.0438
##
## sigma^2 = 0.004377:  log likelihood = 147.05
## AIC=-278.1   AICc=-276.88   BIC=-255.35
```

Novamente, nem todos os coeficientes são significativos.

Agora, testaremos com outros modelos candidatos, no caso ARIMA(0,0,3)(2,1,0), que também obteve um critério AICC relativamente baixo.

```
mod_dummy1 = Arima(arrecad_ts, xreg=covid1, order=c(0,0,3),
                     seasonal=c(2,1,0),include.drift=T,
                     lambda=0)
list(mod_dummy1)
```

```
## [[1]]
## Series: arrecad_ts
## Regression with ARIMA(0,0,3)(2,1,0)[12] errors
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ma1     ma2     ma3     sar1     sar2  drift     xreg
##       0.5687  0.4479  0.3221  -0.6423  -0.4078  5e-03  -0.0202
## s.e.  0.0908  0.0961  0.0999   0.0942   0.1014  8e-04   0.0385
##
## sigma^2 = 0.005783:  log likelihood = 132.93
## AIC=-249.87   AICc=-248.51   BIC=-227.91
```

```
mod_dummy2 = Arima(arrecad_ts, xreg=covid2, order=c(0,0,3),
                     seasonal=c(2,1,0),include.drift=T,
                     lambda=0)
list(mod_dummy2)
```

```
## [[1]]
## Series: arrecad_ts
## Regression with ARIMA(0,0,3)(2,1,0)[12] errors
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ma1     ma2     ma3     sar1     sar2   drift     xreg
##       0.5803  0.4954  0.3763  -0.6480  -0.4045  0.0049  -0.0620
## s.e.  0.0867  0.0965  0.1074   0.0935   0.1008  0.0007   0.0542
##
## sigma^2 = 0.005725:  log likelihood = 133.42
## AIC=-250.85   AICc=-249.49   BIC=-228.89
```

```r
mod_dummy3 = Arima(arrecad_ts, xreg=covid3, order=c(0,0,3),
                     seasonal=c(2,1,0),include.drift=T,
                     lambda=0)
list(mod_dummy3)
```

```
## [[1]]
## Series: arrecad_ts
## Regression with ARIMA(0,0,3)(2,1,0)[12] errors
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ma1     ma2     ma3     sar1     sar2   drift    xreg
##       0.5749  0.4456  0.3247  -0.6391  -0.4131  0.0048  0.0376
## s.e.  0.0900  0.0962  0.0997   0.0943   0.1005  0.0007  0.0432
##
## sigma^2 = 0.005755:  log likelihood = 133.17
## AIC=-250.34   AICc=-248.98   BIC=-228.38
```

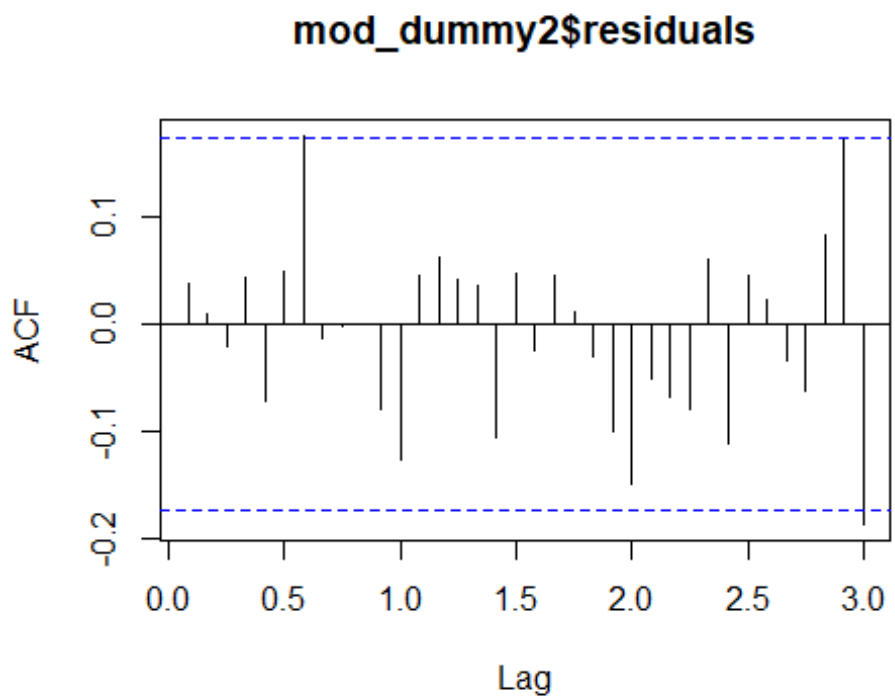
```r
mod_dummy4 = Arima(arrecad_ts, xreg=covid4, order=c(0,0,3),
                     seasonal=c(2,1,0),include.drift=T,
                     lambda=0)

list(mod_dummy4)
```

```
## [[1]]
## Series: arrecad_ts
## Regression with ARIMA(0,0,3)(2,1,0)[12] errors
## Box Cox transformation: lambda= 0
##
## Coefficients:
##          ma1     ma2     ma3     sar1     sar2  drift     xreg
##       0.5811  0.4862  0.3811  -0.6513  -0.4041  5e-03  -0.0639
## s.e.  0.0868  0.0948  0.1044   0.0936   0.1011  7e-04   0.0500
##
## sigma^2 = 0.005705:  log likelihood = 133.61
## AIC=-251.23   AICc=-249.87   BIC=-229.27
```

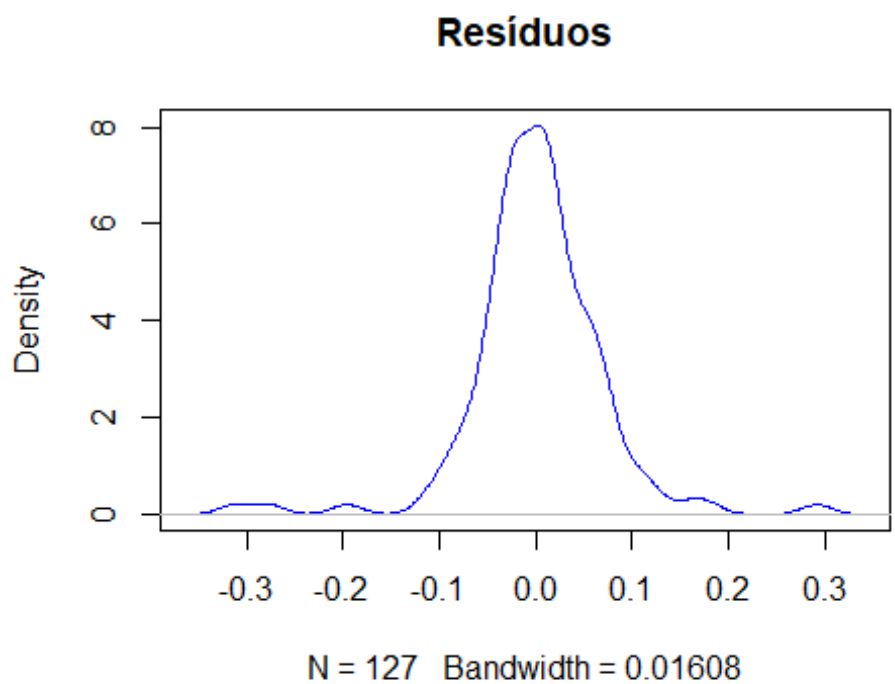
O modelo mod_dummy2 obteve os melhores coeficientes.

Testes dos Resíduos

```r
# Função de autocorrelação dos resíduos com dummy (FAC)
Acf(mod_dummy2$residuals, lag.max=36)
```




```
plot(density(mod_dummy2$residuals,
             kernel = c("gaussian")),
     main="Resíduos", col="blue")
```



```r
Box.test(mod_dummy2$residuals,lag=8,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  mod_dummy2$residuals
## X-squared = 5.7633, df = 3, p-value = 0.1237

Box.test(mod_dummy2$residuals,lag=10,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  mod_dummy2$residuals
## X-squared = 5.764, df = 5, p-value = 0.3299

Box.test(mod_dummy2$residuals,lag=12,
type="Ljung-Box", fitdf=5)

##
##  Box-Ljung test
##
## data:  mod_dummy2$residuals
## X-squared = 8.9364, df = 7, p-value = 0.2573

# Teste de Heterocedasticidade
ArchTest(mod_dummy2$residuals)

##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  mod_dummy2$residuals
## Chi-squared = 20.324, df = 12, p-value = 0.0612

# Teste de Normalidade
shapiro.test(mod_dummy2$residuals)

##
##  Shapiro-Wilk normality test
##
## data:  mod_dummy2$residuals
## W = 0.88725, p-value = 2.307e-08

# Análise visual
par(mfrow=c(2,2))
hist(mod_dummy2$residuals, freq=F,
ylab='Densidade', xlab='Resíduos', main='Resíduos', col="yellow")

#Função de densidade estimada
qqnorm(mod_dummy2$residuals,
```

```
       ylab='Quantis amostrais', xlab='Quantis teóricos', main='Quantil-
Quantil', col="green")
```

O teste de autocorrelação dos resíduos Ljung-Box, não apresentou evidências significativas de autocorrelação serial nos resíduos. Os valores-p obtidos para os diferentes lags (8, 10 e 12) foram 0.1237, 0.3299 e 0.2573, respectivamente, implicando que não há autocorrelação dos resíduos. Além disso, o teste de heterocedasticidade mostrou um p-valor de 0.0612, indicando a ausência de heterocedasticidade significativa nos resíduos. Isso sugere que a variância dos resíduos não varia de forma sistemática ao longo do tempo. No entanto, o teste de normalidade, realizado por meio do teste de Shapiro-Wilk, revelou um p-valor extremamente baixo de 2.307e-08. Isso indica que os resíduos não seguem uma distribuição normal.

Conclusão

Ao analisar os resultados do melhor modelo estimado (`ARIMA(0,0,3)(2,1,0)`) após várias tentativas, não podemos afirmar que houve um efeito significativo da pandemia de Covid-19 sobre a arrecadação federal. Essa conclusão pode ser tomada graças à baixa significância estatística do coeficiente xreg, mesmo no melhor modelo. Outras abordagens ou modelos mais complexos podem ser necessários para uma análise mais completa e precisa do efeito do isolamento social na arrecadação federal, como uma análise de intervenção multivariada.